User Studies

Formative ("How you design it?")

- Improvement-oriented
- (Paper) prototyping, sketching
- Interviews, surveys, diaries, case studies
- Focus groups, user role playing, personas

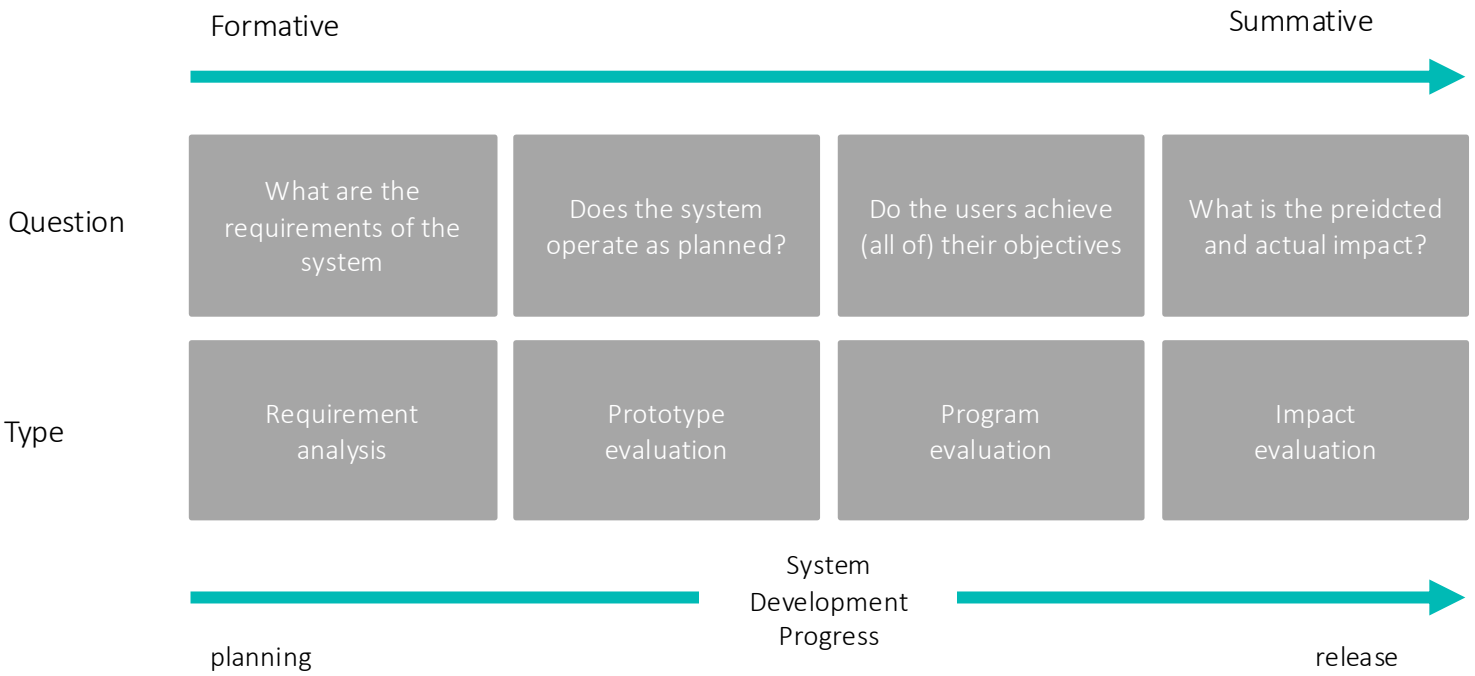
Studies to find out about the next design step.

Summative ("How good is it?")

- based on criteria
- Ratings, questionnaires, scales
- Efficiency measures
- Physiological measures
- …depends on the progress of the development!

Studies to define the degree to which the system fulfils the objectives.

System Evaluation Process



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Orthogonality of Approaches

Methods	Formative	Summative
Analytics	• Cognitive walkthrough • Task analysis	• Heuristic evaluation • Standards compliance
Empirical	• User group identification • Prototype user study	• Usability/UX testing • Field studies



Data	Quantitative	Qualitative
Subjective	• Questionnaires • Visual assessment scales	• Verbal statements • Textual feedback
Objective	• Task completion time • Physiological measures	• Grounded Theory

User Studies – Literature Review

- “Creation of knowledge on the basis of previous research…”
- Analytical study of existing research
- Compare studies; gather and understand all knowledge on a specific topic
  - How have problems in this subject area been solved so far?
  - Analysis of strengths and weaknesses of previous solutions (uncertainty factors)
  - Identification of “best practices” (but also problems)
- Critical assessment of the quality of a methodology
- Different fields/domains have different variables and priorities
- Real vs. assumed knowledge – systematic review results in good overview
- Put own work in context and know what others have already done on this...

Formative Studies – Analytical Methods

Formative analytical methods **without users** can be divided into

- Task analysis
- Expert evaluation

are intended **to complement, not replace, user testing**

- e.g., to avoid gross errors in advance
- User studies are expensive and possibly impossible in early phases (e.g., in a design
- phase without a prototype...)
- To use the more extensive user tests in a more targeted way and not to get invalid results in detail, because early flaws could also be detected by experts...

Expert Review

- Use a small number of reviewers (developers, team members,…)
- Conduct an informal or guideline-based review
  - Consistency check
  - Get indications and hints
  - Identifiy minor and major problems
- Qualitative
  - Observe user interactions (video, screen recordings,…)
  - User explanations and opinions (audio)
  - Anecdotes, transcripts, problem areas, …
- Quantitative
  - Logs, user actions, speed, error rate, …



Slide based on: Valentin Schwind, Evaluation & Experiments, CC BY-SA 4.0

Cognitive Walkthrough

- A **formative analytical evaluation** and **simulation process** that takes a list of questions surveying experts while completing tasks
- The designer (or design team) specifies and (successfully performs) a **series of tasks** on which one will evaluate the design
- **One or more experts go through a problem-solving** and feedback evaluation processes
  - If an evaluator expects no problems at a given step, that judgment must be defended
  - If problems are expected, they should be described

Lewis et al. 1990. Testing a walkthrough methodology for theory-based design of walk-up-and-use interfaces. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '90)

Cognitive Walkthrough

Main purpose

- Focus on problems potential users might have when they use the product for the first time
- Cannot assess learning effects (how fast will a novice user become an expert user)
- Most effective when experts are presented a final design
- Emphasis on future users, as well as their aims and experience: “can this design lead a novice user to the correct solution?”



Cognitive Walkthrough – Basic Procedure

Defining the input

- Who are the users of the system?
- What are the users' goals (and correct actions)?
- What task(s) will be analyzed?
- What action(s) are required for each task?

Conducting the walkthrough

- Will the users try to achieve the right effect?
- Will the user notice that the correct action is available?
- Will the user associate the correct action with the effect to be achieved?
- If the correct action is performed, will the user see that progress is being made toward solution of the task?

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Cognitive Walkthrough – Example Template

EVALUATOR _____ DATE _____

INTERFACE _____ TASK _____ STEP #_____

1. Description of user's immediate goal: _____
2. (First/next) atomic action user should take: _____
   2a. Obvious that action is available? Why/why not?
   2b. Obvious that action is appropriate to goal? Why/Why not?
3. How will user access description of action?
   3a. Problem accessing? Why/Why not?
4. How will user associate description with action?
   4a. Problem associating? Why/why not?
5. All other available actions less appropriate? For each, why/why not?
6. How will user execute the action?
   6a. Problems? Why/why not?
7. If timeouts, time for user to decide before timeout? Why/why not?
8. Execute the action. Describe system response: _____
   8a. Obvious progress has been made toward goal? Why/why not?
   8b. User can access needed info. in system response? Why/why not?
9. Describe appropriate modified goal, if any: _____
   9a. Obvious that goal should change? Why/why not?
   9b. If task completed, is it obvious? Why/why not?

Slide based on: Valentin Schwind,
Evaluation & Experiments, CC BY-SA 4.0

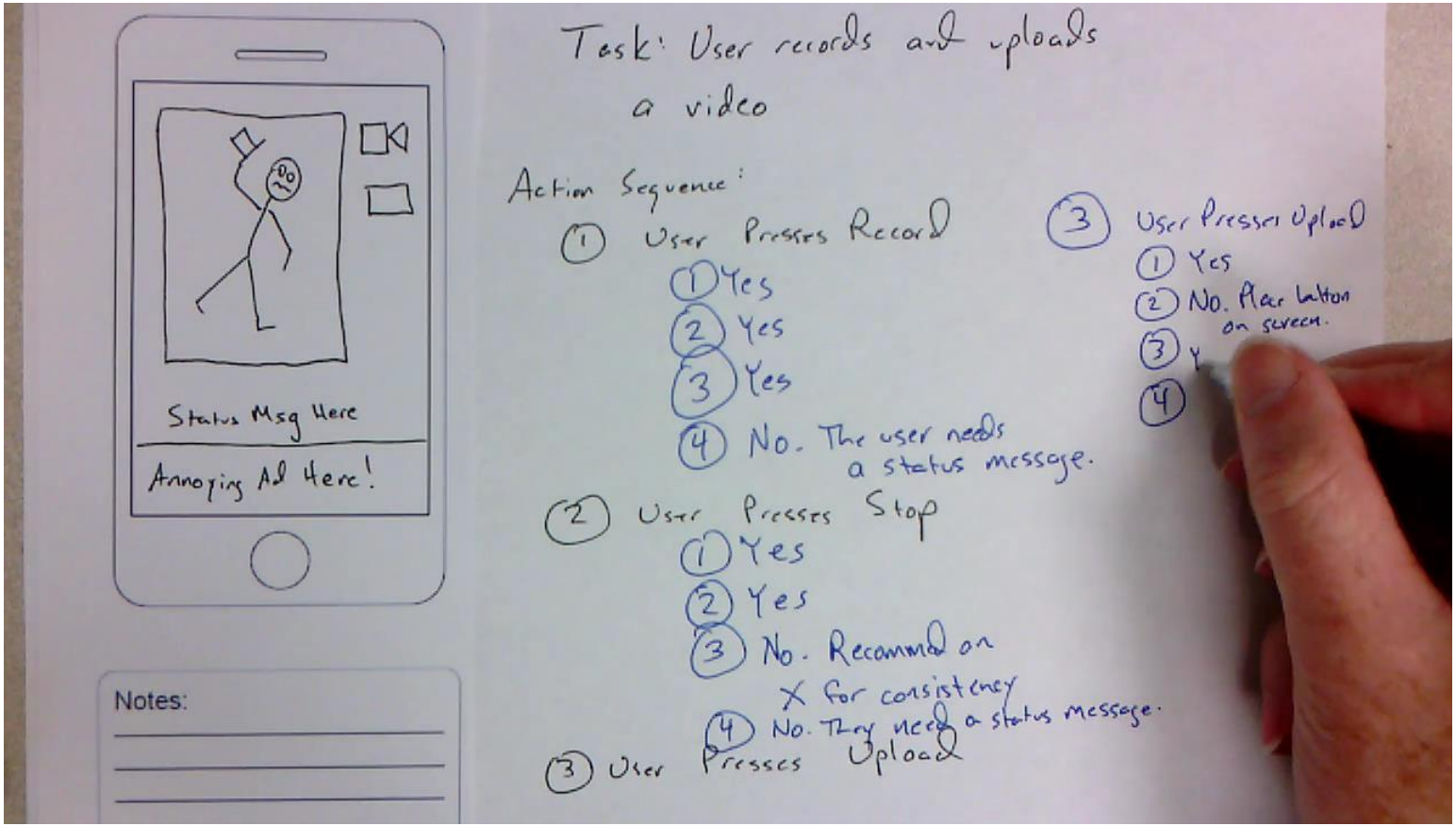
Cognitive Walkthrough – Example

Task: „Upload a video on the website“

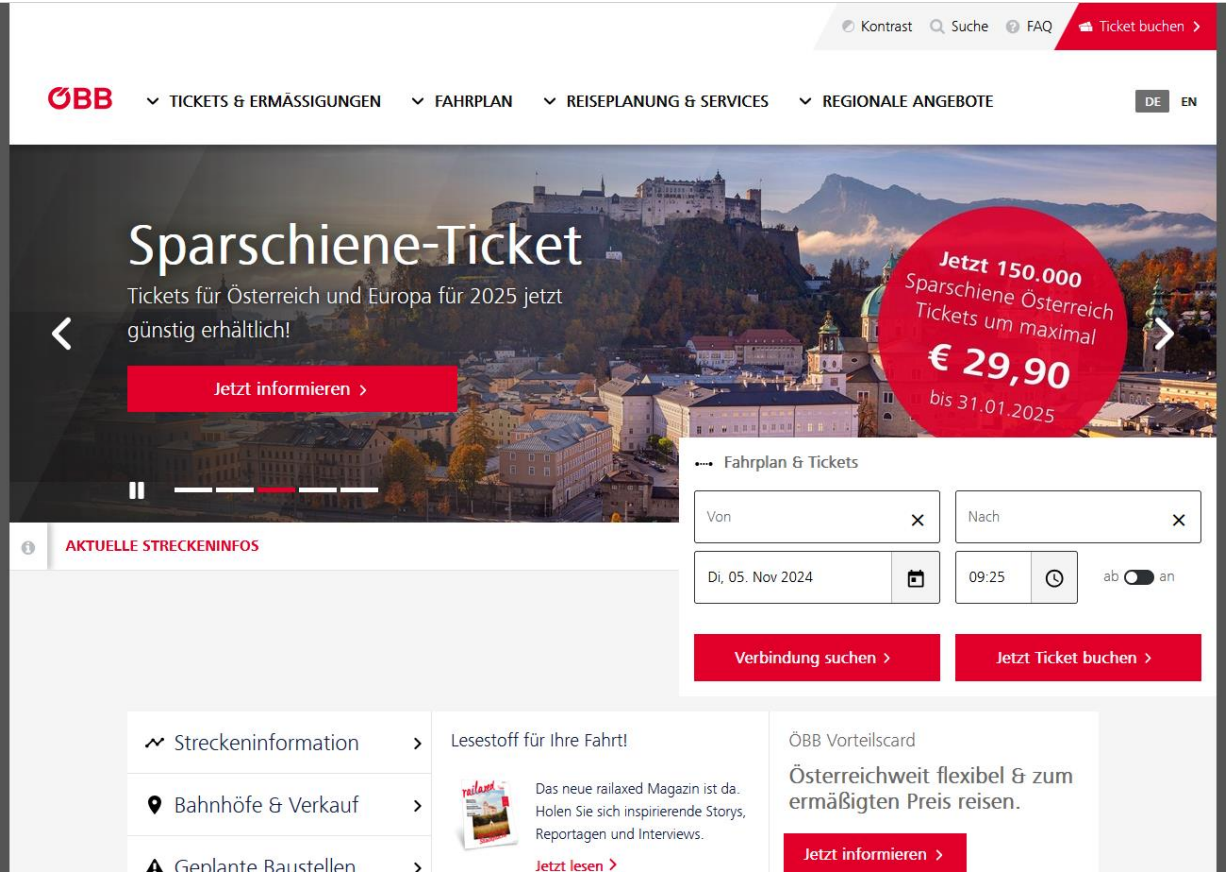
- Action 1: „Click the upload button“ – visible?
- Action 2: „Enter a file title“ – clear why?
- Action 3: „Enter a description“ – required?
- Action 4: „Find the file on your system“ – what about links?
- Action 5: „Start upload“– button visible/disable/enabled?
- Action 6: „Waiting for upload“ – clear visual feedback?
- Action 7: „Confirming upload“ – video playback?
- …

Slide based on: Valentin Schwind, Evaluation & Experiments, CC BY-SA 4.0

Cognitive Walkthrough – Example



Cognitive Walkthrough – Example



Cognitive Walkthrough – Questions to ask

- Will the user try to achieve the desired effect
- Will the user clearly see what to do and how?
- Will the user recognize which action will lead to the desired effect?
- Will the user understand the feedback?



Cognitive Walkthrough – Typical user problems

"Failure stories" can usually be categorized to the major questions

- The user does not try to achieve the desired effect
  - Can the action be automated?
  - Can a tooltip or help text indicate the necessary actions?
  - Can the task be restructured so that the necessary actions become visible to the user?
- The user does not recognize that the correct action can be performed
  - Are there better controls (i.e., selection instead of text input field)?
- The user cannot recognize that a given action will lead to the desired effect
  - Can the arrangement, design, or labeling of the elements be improved?
- The user cannot perceive the process initiated by an action
  - How can feedback be improved?

Cognitive Walkthrough

Advantages

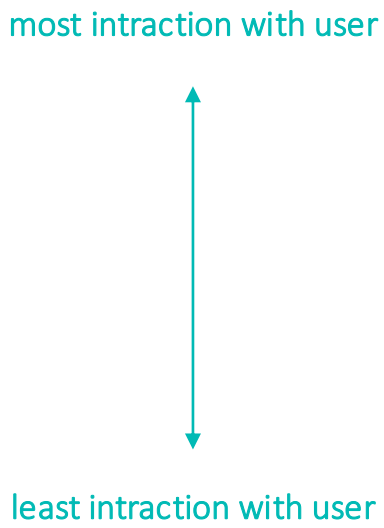
- Flexible, quick, and easy to do
- Can be used in early development stages (e.g., low-fidelity, paper prototypes)
- Errors recognizable in the approach
- Makes explicit the decisions that have been made in the process of designing an interface

Disadvantages

- Artificial setting and tasks
- Realistic scenarios not guaranteed
- Evaluator cannot objectively assess the user
- Inflexible in advanced development stages
- Not all problems can be revealed (e.g. 15 of 18)
- No user experience sampling (emotions, satisfaction,…)
- Not suitable for complex systems

User Research

- **Interviews:** one-to-one interaction between the user and researcher
- **Focus groups:** facilitating a group discussion
- **Think aloud:** users talking through thoughts while interacting with a system or solving a problem
- **Ethnographic studies:** understanding how humans use technology in their daily lives
- **Observations:** observing people working and interacting with technologies
- **Surveys:** questionnaires can collect information from far more people
- **Cultural Probing:** Self-observation where direct observation is not possible or too intrusive
- **Diary studies:** participants maintain a diary of relevant actions, experiences or thoughts



Ann Blandford; Dominic Furniss; Stephann Makri, Qualitative HCI Research: Going Behind the Scenes, Morgan & Claypool, 2016



Quelle: Steve Mulder http://www.muldermedia.com/

Choosing the methods

A proper method(s) must be selected depending on the focus

- Usability, User Experience, etc.
- Time effort, scope, level of detail
- Quantitative vs. qualitative methods
- Lab vs. field studies
- Costs, etc.

There is not perfect set of methods – it entirely depends on the context

Every method has its “traps” that can lead to misleading results

- Too many instructions can influence interviews/discussions
- Wrong selection of participant sample, etc.

INTERVIEWS

Interviews

WHY (goals): gathering people's perceptions and experiences

WHAT (data gathered):

- qualitative: handwritten notes (open questions)
- quantitative: short survey at the beginning, closed questions

Interview types

- *unstructured:* exploratory and similar to conversations
- *structured:* predetermined set of (ordered) questions
- *semi-structured:* guided by a script but can become more open as they progress
- *group (aka Fokus groups):* enable diverse views



Ann Blandford; Dominic Furniss; Stephann Makri, Qualitative HCI Research: Going Behind the Scenes, Morgan & Claypool, 2016

Interviews

Question types:

- *Closed questions:* have a predetermined answer, e.g. „yes“, „no“
- *Open questions:* no predetermined format, e.g. „Why do you use this device?“ (generally better for gathering information on users needs)

Information needed:

- **the context:** when, why, and how the product (or analogous system) is or will be used
- **domain knowledge:** What do users need to know to do their jobs?
- **current tasks and activities:** both those the current product is required to accomplish and those it doesn’t support
- **goals and motivations** for using their product
- **problems and frustrations** with current products

Ann Blandford; Dominic Furniss; Stephann Makri, Qualitative HCI Research: Going Behind the Scenes, Morgan & Claypool, 2016

Interviews

Advantages

- Every participant has enough time to express his/her opinion
- Reveal in-depth information
- Complex questions/issues can be clarified

Disadvantages

- Needs a lot of time
- Requires resources for evaluation, documentation
- Not representative as usually just a few participants are interviewed



Interviewing Skills

How to avoid bias caused by the interviewer?

- Don’t ask questions that are too restrictive. You might end up with a series of yes/no answers otherwise.
- Ask the interviewer to qualify their responses (if vague), but be careful:
  - Good: “What do you mean by that?”
  - Bad: “Do you mean this …?”

- Never reveal your intentions or your opinion on the topic. This might result in so-called interviewer effects.
  - Good: “Please name 3 positive and 3 negative aspects about this application.”
  - Bad: “This is my final year project for my bachelor's degree. I spent a lot of time programming the application. How do you like it?”



Interviewing Skills

- Don’t let participants drift off topic.
  - (But at the same time) be flexible about their answers.
- Don’t restrict participants to answering your questions in the order you specified.
- Remember that there are **no right or wrong answers,** only the **interviewee’s opinions**!



FOCUS GROUPS

Focus Groups

- group interview that involves normally 3-10 people, led by a trained facilitator
  - single or multiple sessions
  - open atmosphere (not too structured)
  - can be recorded for later analysis

- participants are selected to provide a representative sample of the target population

WHY (goals): collecting multiple viewpoints (suggestions, impressions, perceptions, opinions, etc.)

WHAT (data gathered):

- qualitative: handwritten notes: open questions
- quantitative: e.g. short survey at the beginning, counted opinions on different topics



Ann Blandford; Dominic Furniss; Stephann Makri, Qualitative HCI Research: Going Behind the Scenes, Morgan & Claypool, 2016

Focus Groups

Principle: “The user is the expert; the researcher listens without contributing a personal opinion”

Focus Group moderation

- Be objective, do not influence participants with own opinion
- Provide guidance to achieve your goals
- Step back and let the participants talk
- Step in and mediate when
  - Participants “follow the herd” (i.e., do not express their own opinion)
  - The discussion loses focus or gets out of track (time, topic, etc.) → Attention: this can introduce a bias!

- Emphasize that it is about the participants’ needs and by no means only about what is technically feasible!



Focus Groups

Advantages

- Ideas are controversially discussed which generate new ideas
- Faster than many interviews
- Additional insights which cannot be assessed in interviews, etc.

Disadvantages

- Can be dominated by just a few participants
- Some participants might just follow the strong opinions
- Not representative (not all users can be included)



ETHNOGRAPHIC STUDIES

Ethnographic Studies

"Traditional" ethnographic observations

Method from social anthropology:

- researchers spend time (weeks, months, even years) in another culture
- This allows to study cultural differences to obtain a better understanding

Ethnographic observation in HCI

- Aims at understanding how humans use technology in their daily lives
- Ethnographic study ("immerse in a culture") = living the life of (potential) users
  - Active participation allows to gain (otherwise potentially hidden) insights
  - Interpretation may require "insider information"
  - No hypotheses needed!

- Example: To develop an application for waiters in bars/restaurants, it might be useful to participate (or even work as waiter) to know about the problems/workflows

Ethnographic Studies

"To immerse in a culture" can involve

- **Pure observation:** Directly/indirectly observing what happens in specific settings
- **Ethnographic interviews:** Asking questions and conversing with humans in specific settings to derive useful information
- **Documentation:** Writing a report about what has been observed



Ethnographic Studies

Types of ethnography

- **Parallel:** Ethnographic studies are conducted simultaneously with product development
- **“Quick and Dirty”:** Getting quick but qualified insights into the product environment or a specific setting
- **Evaluative:** Testing the validity of formulated design requirements/decisions



Ethnographic Studies - Example

"One Laptop per Child" project

- Aims at providing an affordable computer and software for education in developing countries

- Ethnographic studies in target cultures/countries
  - Defining a child-oriented and attractive design (motivation)
  - Understanding of education systems and traditions, including relevant stakeholders (organizations, authorities)
  - Identification of potential prejudices (against the project)
  - Identification of potential downsides
  - Does the target group really benefit from the project?

- Ethnographic studies were conducted (maybe not in the necessary extent)





Bernhard Preim, Raimund Dachselt, „Interaktive Systeme – Band 2", 2. Auflage, Springer Vieweg, 2015

OBSERVATION

Observations

help designers understand the users' context, tasks, and goals

WHY (goals): to watch the users in action in their own environment

WHAT (data gathered):

- mostly qualitative: handwritten notes, audio, video, images
- some quantitative: short survey at the beginning, closed questions

most commonly take place "in the wild," but also possible in a controlled environment

Observation types:

- semi-structured, informal (especially at the beginning of the project)
- structured, systematic observation (supported by worksheets & checklists)

Ann Blandford; Dominic Furniss; Stephann Makri, Qualitative HCI Research: Going Behind the Scenes, Morgan & Claypool, 2016

Observations

Direct vs. Indirect

- Direct observation: observing objects/processes (i.e., shopping behavior of customers)
- Indirect observation: observation of potential effects only (i.e., comparison of sales figures)

Field vs. Lab

- Field observation: Investigation of real natural context (i.e., in a shop during opening hours)
- Lab observation: Investigation in an artificial setting (i.e., evaluating different presentations of goods in a controlled lab environment)



Observations

Systematic and objective assessment requires to define a process

- Who will observe what, where, and when?
- How are results/insights recorded?
- How will the results be interpreted?

Data logging techniques

- Photos, Videos
- Audio recordings
- “Tracking”
  - Logging of user behavior (i.e., button/keys used in an application)
  - Extra devices for logging (movement patterns using GPS, etc.)
  - “Motion Capturing”, etc.



Observations

- Preparation (Piloting the whole process at least once before the actual start)
  - Understanding of system/objects/processes (cultural, organizational aspects, etc.)
  - Formulation of the goals and preparation of important questions
  - Getting permission for data logging (fotos, videos, etc.)

- Field study
  - Acquisition of participants
  - Direct or indirect observation, interviews, in-situ collection of information

- Analysis
  - Data analysis/aggregation, quantification, summary of results
  - Interpretation of findings
  - Refining goals, problems (may lead to another observation)

- Documentation and Reporting
  - Of findings (may require multiple documents for different stakeholders and target groups)
  - If possible, record everything you can during the study (provided permission)

THINK ALOUD

Think-Aloud

involves the users of an interactive system articulating their thoughts as they work with that system

typically focus on the interaction with a particular interface

WHY (goals): understanding how people perceive and experience system, and how they use it to support their work

WHAT (data gathered):

- mostly qualitative: handwritten notes, audio
- some quantitative: e.g. counts of various actions



Ann Blandford; Dominic Furniss; Stephann Makri, Qualitative HCI Research: Going Behind the Scenes, Morgan & Claypool, 2016

SURVEY

Survey

- well-established technique for collecting demographic data and users' opinions, experience and needs
  - **Analysis Phase:** identify user needs or reactions to product ideas
  - **Evaluation Phase:** evaluate different designs/prototypes
- can be distributed to a large number of participants

WHY (goals): answering specific questions, collecting demographic information, users' opinions



H. Sharp., J. Preece, Y. Rogers.: Interaction Design - beyond Human-Computer Interaction, Wiley; 5th Edition, 2019

Survey

WHAT (data gathered):

- quantitative: closed ended questions: ordered response (e.g. Likert scales, Rank), unordered response (e.g. gender or education level)
- qualitative: open-ended questions (e.g. opinions)

Using a **standardized questionnaire** allows for a quantitative evaluation, e.g.:

- **AttrakDiff** (distinguishes between objective usability and subjective user experience)
- **UEQ** (User Experience Questionnaire)
- **SUS** (System Usability Scale)



H. Sharp., J. Preece, Y. Rogers.: Interaction Design - beyond Human-Computer Interaction, Wiley; 5th Edition, 2019

System Usability Scale (SUS)

This is a standard questionnaire that measures the overall usability of a system. Please select the answer that best expresses how you feel about each statement after using the website today.

		Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree
1.	I think I would like to use this tool frequently.	☐	☐	☐	☐	☐
2.	I found the tool unnecessarily complex.	☐	☐	☐	☐	☐
3.	I thought the tool was easy to use.	☐	☐	☐	☐	☐
4.	I think that I would need the support of a technical person to be able to use this system.	☐	☐	☐	☐	☐
5.	I found the various functions in this tool were well integrated.	☐	☐	☐	☐	☐
6.	I thought there was too much inconsistency in this tool.	☐	☐	☐	☐	☐
7.	I would imagine that most people would learn to use this tool very quickly.	☐	☐	☐	☐	☐
8.	I found the tool very cumbersome to use.	☐	☐	☐	☐	☐
9.	I felt very confident using the tool.	☐	☐	☐	☐	☐
10.	I needed to learn a lot of things before I could get going with this tool.	☐	☐	☐	☐	☐

How likely are you to recommend this website to others? (please circle your answer)

Not at all likely 0 1 2 3 4 5 6 7 8 9 10 Extremely likely

Please assess the product now by ticking one circle per line.

	1	2	3	4	5	6	7		
annoying	○	○	○	○	○	○	○	enjoyable	1
not understandable	○	○	○	○	○	○	○	understandable	2
creative	○	○	○	○	○	○	○	dull	3
easy to learn	○	○	○	○	○	○	○	difficult to learn	4
valuable	○	○	○	○	○	○	○	inferior	5
boring	○	○	○	○	○	○	○	exciting	6
not interesting	○	○	○	○	○	○	○	interesting	7
unpredictable	○	○	○	○	○	○	○	predictable	8
fast	○	○	○	○	○	○	○	slow	9
inventive	○	○	○	○	○	○	○	conventional	10
obstructive	○	○	○	○	○	○	○	supportive	11
good	○	○	○	○	○	○	○	bad	12
complicated	○	○	○	○	○	○	○	easy	13
unlikable	○	○	○	○	○	○	○	pleasing	14
usual	○	○	○	○	○	○	○	leading edge	15
unpleasant	○	○	○	○	○	○	○	pleasant	16
secure	○	○	○	○	○	○	○	not secure	17
motivating	○	○	○	○	○	○	○	demotivating	18
meets expectations	○	○	○	○	○	○	○	does not meet expectations	19
inefficient	○	○	○	○	○	○	○	efficient	20
clear	○	○	○	○	○	○	○	confusing	21
impractical	○	○	○	○	○	○	○	practical	22
organized	○	○	○	○	○	○	○	cluttered	23
attractive	○	○	○	○	○	○	○	unattractive	24
friendly	○	○	○	○	○	○	○	unfriendly	25
conservative	○	○	○	○	○	○	○	innovative	26

Survey

- Use **standard questionnaires** to reduce the effort required

- For practical applications, your own individual questionnaires are often sufficient
  - But(!) your **results may not be reliable**
  - Still, problems can be detected

- Questionnaires are **suitable for collecting both qualitative and quantitative information** (an analytical evaluation with questionnaires is also possible (cf. guidelines inspection))



Survey

Advantages

- Many participants can be included (compared to interviews, lab studies, etc.)
- Online: fast, automated evaluation, anonymity

Disadvantages

- No follow up questions or discussions
- Potential misunderstandings when no experimenter can clarify
- Often difficult to assess specific problems (especially in the early phase)
- Online surveys: hard to assess users with low technology affinity



Survey Creation

Creation of a questionnaire **according to scientific criteria** is quite complex

- Selection and definition of the constructs to be measured
- Formulation and selection of items
- Standardization and calibration with large sample (N > 300)

A questionnaire **needs to be evaluated** itself before it can be used

- **Objectivity:** Results independent of the investigator
- **Reliability:** Does the questionnaire provide approximately the same results when the test is repeated?
- **Validity:** Does the questionnaire measure what it is supposed to measure?



Before implementation

What is the evaluation going to cost? Don't forget hidden cost, e.g.,

- Invitation
- Telephone costs (telephone interviews)
- Printout of questionnaire (paper+toner)
- Costs for preliminary studies
- Data analysis costs
- Administrative costs
- Participation fee for test participants
- and so on.



Before implementation

What do I want to collect/measure/find out with the questionnaire?

- About the user
  - settings, preferences, wishes, behavior, previous experience, knowledge
- About the product/application/interface
  - Inconsistencies in use, missing "controls„ (e.g., back button, undo function), misbehavior/malfunction, lack of functionality



Before implementation

Why do I need (or want) to know this?

- To be able to formulate/test hypotheses
- To be able to define follow-up actions
- To be able to develop more specific, detailed questions
- To justify the design decision
- To be able to make a choice between System X and Y
- To be able to define the basic behavior of a program for evaluation



Before implementation

What will I do with the results?

- Who will use the data/results?
- In what way?
- How/to whom will I report (conspicuous) data?



Before implementation

Do I even know enough about the topic to be able to define meaningful questions? / Where can I get the necessary information?

- Literature
- Experts
- Function or task analysis
- Inspection of questionnaires from similar situations
- Historical data
- Interviews



Before implementation

In summary, the **most important questions** to answer during planning are

- Which questions should be answered?
- How much time do the subjects have for the survey?
- How should the data evaluation be done afterwards?

It must be clear **what results the survey should deliver**, before the study → impact on design and collected data!

The content and design of the questionnaire is derived from this



Survey Content

- **Demographic information/personal data**, e.g., age, gender, education, previous knowledge/experience, individual limitations

- Questions about the **overall system or individual use cases**

- Questions about **various aspects**, depending on the application, e.g.,
  - Understandable or confusing
  - Effective or cumbersome
  - Inaccurate or precise
  - Misleading or helpful
  - Dissuasive or encouraging
  - …



Survey – Good Questions

The "doctrine of the question" (from a social science point of view) focuses on the following problem areas:

- **Why** is the question asked?
- **How** should the question be formulated?
- **What** are the rules?
- **What kind** of question (and answer) is appropriate?
- **Which** sample should be selected?



Survey – Good Questions

Short, simple and related to the respondent's frame of reference (i.e. using the "user language" and not foreign words)

- Questions should not simply be freely formulated
- Questions that are too general are misleading - beware of multidimensionality
- No double negation or unclear words
- Distorted questions, i.e. not suggestive, hypothetical or tendentious (e.g. not asking hypotheses as direct questions) but neutral

Add Explanations and Descriptions



Survey – Good Questions

Closed questions

- Answer specifications such as scales ("Likert-Scale"), multiple-choice, ranking, no answer/"don't know“

Open questions (free text)

- Descriptions
- Opinions
- Reasons

- Rule of thumb: The more prior knowledge, the more closed questions

Other kinds

- Hybrid questions
- Filter questions
- Main and follow-up questions
- Control questions
- Concluding questions

Survey – Good Questions

- Use neutral questions; no questions that implicitly influence the answer
- Use generally understandable vocabulary (which can be assumed to be understood by any person)
- For German-language questionnaires: Do not use double questions or questions with double negation!
- A questionnaire should have a logical structure
  - Rule of thumb: The mindfulness of the subjects is often best in the middle third → important or even difficult questions should be asked in the middle third
  - "Funnel principle": Start with the general/"harmless" questions, and ask the more specific questions later
  - "K-I-S-S": Keep it short and simple



Biases and Missunderstandings

Social desirability and approval tendency

- “Social desirability” refers to the tendency of respondents to give answers that they believe are desired by the investigator or society

Sponsoring effect

- The person filling out the questionnaire (or being interviewed) knows who commissioned the study

Framing effect

- Study participants decide differently, depending on the type or wording of the question formulation

Be aware of implicit dependencies ("Primacy and recency effects") Questions can be implicitly influenced by previous questions

CULTURAL PROBING

Cultural Probing

Self-observation where direct observation is not possible or too intrusive

- In private and sensitive environments (i.e., in the bedroom, hospital, etc.)
- Participants receive “probe packages”: those propose concrete tasks/activities to make participants “think”; then participants document processes, ideas, emotions, etc. in a log
- Allows to derive insights into participants’ lives and behaviors
- Based on interpretation, designers can create new ideas (sometimes it is necessary to discuss with participants retrospectively)

Probe-package can consist of cameras, audio recording devices,diaries, post-its, pens and paper, glue, cards, etc.

Cultural Probing

Advantages

- Traditional methods may influence results (“observer bias”), especially in sensitive domains (privacy)
- Traditional methods sometimes focus on what participants do, but not necessarily why and how they feel about something

Disadvantages

- No defined process/standard, requires good study design
- Slow

Results

- Ideas for products, designs, extensions, etc. in the investigated context
- Collected material can inspire designers to improve ideas



DIARY STUDY

Diary studies

Systematic logging of data over a longer time period

"A study that asks people to ***keep a diary****, or journal, of* ***their interactions*** *with a computer system/interface/ device/application.* ***Any significant events or problems*** *during their use, or other aspects of their working life,* ***should be noted down****.*

A diary typically asks a user to ***record the date and time of an event****, where they are, information about the event of significance, and ratings about* ***how they feel****, etc."*



Diary studies

- enable participants to record data in their own time, at particular times of day or when a particular trigger occurs
- show thoughts, feelings or behaviors of users at crucial moments
- logging usually takes place directly, during or immediately after an experience

WHY (goals): to watch the users in action in their own environment

WHAT (data gathered):

- qualitative: handwritten notes, video, audio, images (impressions, experiences, problems, emotions)
- quantitative: e.g. ordered response (user behavior, frequency and duration of use, scales, etc.

Ann Blandford; Dominic Furniss; Stephann Makri, Qualitative HCI Research: Going Behind the Scenes, Morgan & Claypool, 2016



ETHICAL AND LEGAL ASPECTS

Ethics and informed consent

VIP is a useful mnemonic for the main concerns, Vulnerability, Informed consent, and Privacy:

- Vulnerability: particular care needs to be taken when recruiting participants from groups that might be regarded as vulnerable, such as children, the elderly, or people with a particular condition (illness, addiction, etc.)
- Informed consent: where possible, participants should be informed of the purpose of the study, and of their right to withdraw at any time. It is common practice to provide a written information sheet outlining the purpose of the study, what is expected of participants, how their data will be stored and used, and how findings will be reported
- Privacy and confidentiality should be respected, in data gathering, management and reporting

https://www.interaction-design.org/literature/book/the-encyclopedia-of-human-computer-interaction-2nd-ed/semi-structured-qualitative-studies

Ethical and legal aspects

- Are there relevant regulatory requirements? (Laws, regulations, voluntary commitments)

- Patents, utility models, or other potentially secured rights
- Security, privacy aspects
- Ethical, cultural, and religious aspects (i.e., care-robots in elderly care)

- Technology assessment (short- and long-term consequences of technology)

  Example
  - A new warning system informs truck drivers of dangers at right turns (“blind spot” pedestrian safety)
  - If too many false alarms drivers might miss an important warning (“cry wolf effect”)
  - Drivers might reduce own safety measures (mirror glances)
  - How can a potential long-term benefit be evaluated and guaranteed?

USER STUDIES

DESIGN AND EXECUTION

User Studies – Design and Execution

1. Have a clear research goal
2. Involve all the stakeholder groups
3. Create a study design document including
   - your research question
   - participant profile
   - compensation plan
   - your methodology
   - timeline
   - types of data and how you will analyse the data
4. Pilot your study
5. Make it robust enough
6. Use existing technologies
7. Start data interpretation soon after data gathering session
8. Initial interpretation before deeper analysis

Do

Inform participants about…

- general study goals
- study procedure (duration, tasks, etc.)

Get participants' written consent

- tell them they can quit any time
- tell them it is not about their performance, but the one of the system!

Ensure confidentiality and anonymity

Inform on potential risks

Don't

- inform participants about your hypotheses
- tell participants which group they are in
- tell participants the details of the experiments (only the minimal part they absolutely must know)
- fail to obtain permission of an ethics commission when you should apply for one

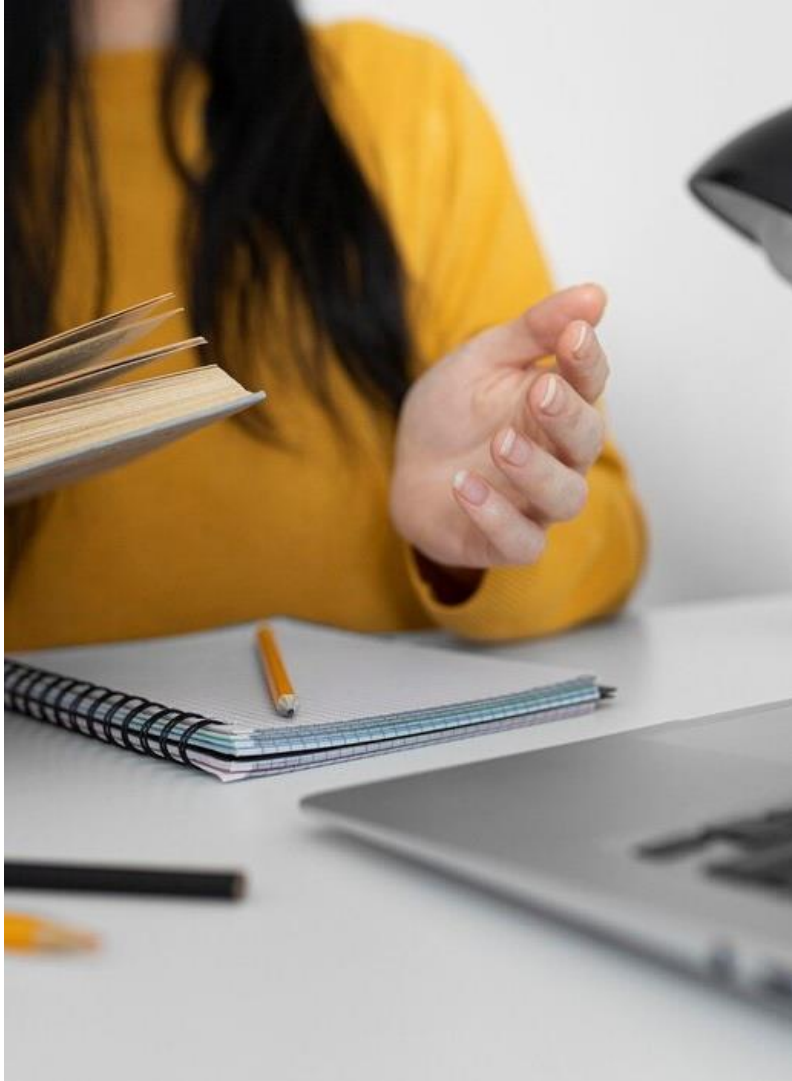
User Studies – Steps

1. Welcome the participant
   - inform the user about the study
2. Sign the consent form
   - answer questions if necessary
3. Pre-survey: A first questionnaire, e.g.,
   - Demographics data
   - questions of interest to the experiment
4. Give the participant instructions on the task
   - only explain one task at once;
   - your explanations must be identical for all participants!
   - Important: do not reveal your hypotheses!
5. Tasks
   - perform the actual run and measure variables
   - (optional) do a survey on subjective measure (post-task)
   - after all tasks have been done: (optional) questionnaire (post-test)
6. Finally
   - Debriefing: thank the participant, provide payment, etc.
   - be available for questions of participants or for their (informal) feedback

User Studies

- Get written consent from participants
- Templates are available, e.g. from “World Health Organization” (WHO)
- May be checked with the legal department/ review board of your University

https://www.who.int/groups/research-ethics-review-committee/guidelines-on-submitting-research-proposals-for-ethics-review/templates-for-informed-consent-forms



CONSENT TO PARTICIPATE IN RESEARCH

Steering behavior control via visual stimuli

You are invited to participate in a research study conducted by Dr. Andreas Riener and Hannes Thaller, from the Department of Pervasive Computing at the Johannes Kepler University. Your participation in this study is entirely voluntary. Please read the information below and ask questions about anything you do not understand, before deciding whether or not to participate. We need drivers who have a driver's license and more than 2 year driving experience. Other than that, there are no exclusion criteria for this study.

• PURPOSE OF THE STUDY

The purpose of this study is to examine the effects of different visual stimuli on the driving performance of driver's. This knowledge will, in turn, help to determine whether visual stimuli could be used to reduce the overall workload during driving. Then, researches can identify how the in-vehicle system would provide adaptive interfaces to drivers to mitigate those information transfers and thereby resulting in better and safer driving performance.

• PROCEDURES

If you volunteer to participate in this study, you will be asked to do the following procedures:

- First of all, you can adjust your seat and the steering wheel for your comfort.
- Then, you will have a test drive session. In the test drive session you will drive on a straight three lane road where you will be asked to change the lane according to a sign. At first all panels on the sign bear a red X which is the default appearance of the panels. As you approach with the car the panels will flush three times switching one panel to a green tile symbol indicating the lane on which you should drive next. Your task is to change the lane as convenient for you but try to be as fast and accurate as possible. After each successful lane change you should stay on the lane till the next sign shows you the next target lane. The completion of a lane change will be signaled by a beep. The entire session will take approximately 10 minutes.
- After that you will fill out a subjective workload questionnaire (*attachment1*).
- At the end of the study, you will be asked to provide some basic information about yourself and driving experience (*attachment2*, e.g. Age, gender, year of driving experience).
- You will be debriefed about the experiment and the meaning of the results.

• POTENTIAL RISKS AND DISCOMFORTS

There are no foreseeable risks or discomforts in this study. The risks involved are no greater than those involved in daily activities such as playing a video game. We believe that they are not likely to happen. If discomforts become a problem, you may discontinue your participation.

• POTENTIAL BENEFITS TO SUBJECTS AND/OR TO SOCIETY

It is not likely that you will benefit directly from participation in this study. However, participation in this study will contribute to our understanding of visual stimuli during driving which subsequently aid our understanding of how to best design the interface for drivers.

• CONFIDENTIALITY

Any information that is obtained in connection with this study and that can be identified with you will remain confidential and will be disclosed only with your permission or as required by law. Confidentiality will be maintained by means of a code number to let Hannes Thaller know who you are. We will not use your name in any of the information we get from this study or in any of the research reports. When the study is finished, we will destroy the list that shows which code number goes with your name.

Information that can identify you individually will not be released to anyone outside the study. We may use any information that we get from this study in any way we think is best for publication or education. Any information we use for publication will not identify you individually.

In case of an emergency, injury, or illness that occurs during this study, I hereby authorize the release of any and all health information to allow for medical care and treatment of my condition.

• PARTICIPATION AND WITHDRAWAL

You can choose whether or not to be in this study. If you volunteer to be in this study, you may withdraw at any time without consequences of any kind. You may also refuse to answer any questions you do not want to answer.

I verify that I am at least age 20. I understand the procedures described above. My questions have been answered to my satisfaction, and I agree to participate in this study. I have been given a copy of this form.

_____
Printed Name of Subject

_____ _____
Signature of Subject Date